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LAB 4: Find Road Signs and Traffic Lights!

Computer Vision 2024-25

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CV 2024/25: HW / Projects

Introduction to Python/OpenCV — No HW

1. Histogram eq. and Filtering
2. Road Line Detection

- Select 1 among LAB1 and LAB2
- First «simple» HW: 2 points
- Provide a very short report or place comments in the source
- On/off mark
- Deadline 29/11

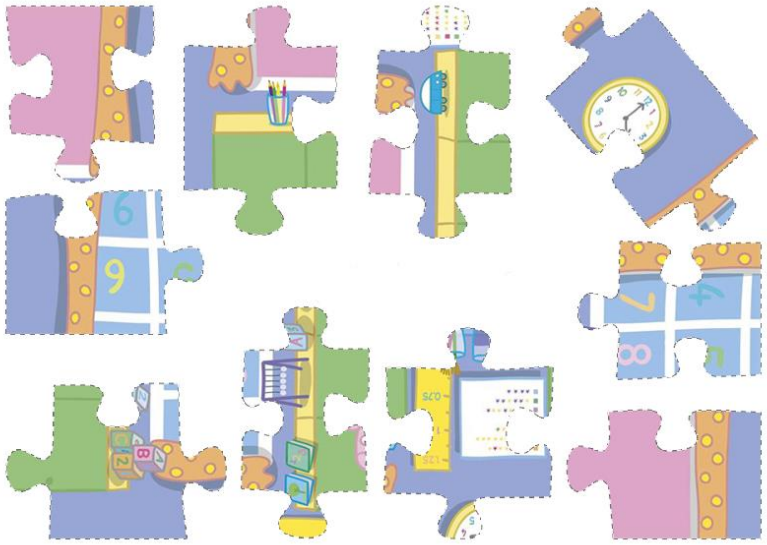
3. Features Matching
4. **Deep Learning**

- Select 1 among LAB3 and LAB4
- Second «advanced» HW: 5 points
- More detailed report with results
- Mark based on solution quality

Final Exam (written in classroom): ~25 points

Final mark: 2 + **5** + 25 = max 32 points

CV 2024/25: Final Projects

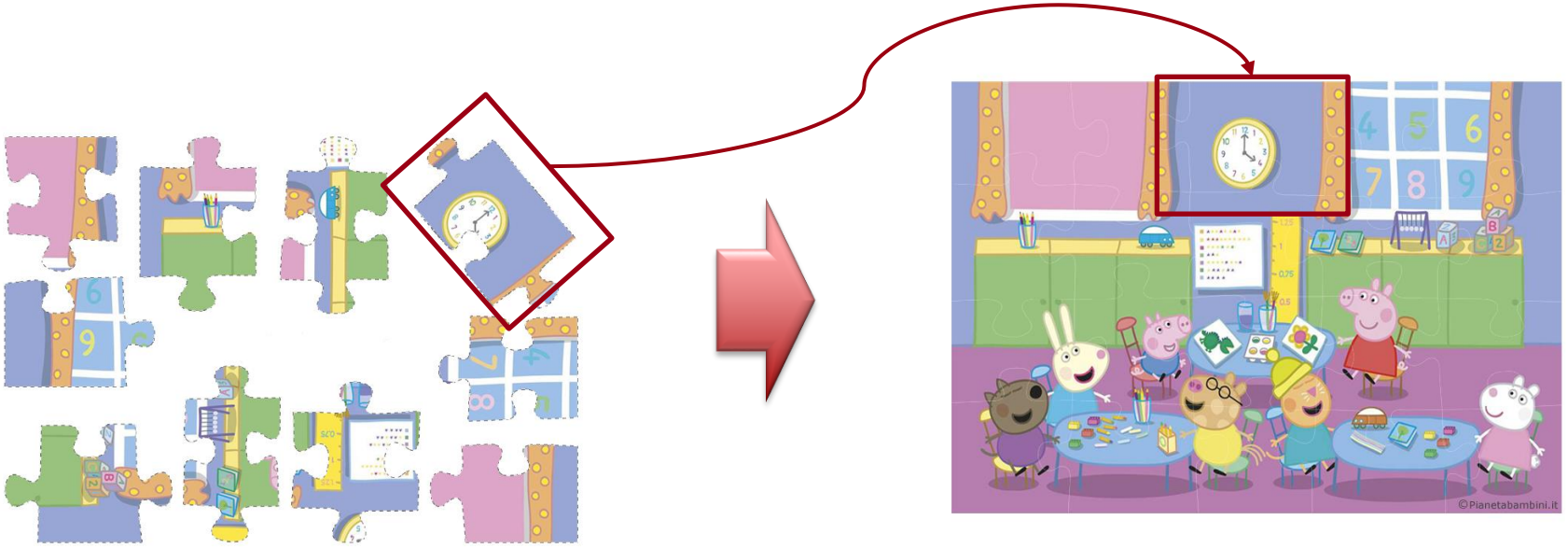




Two options:

1. *Solve the Puzzle (LAB3)*
2. **Detect traffic light and road signs (*LAB4*)**

Project 1: Solve the Puzzle

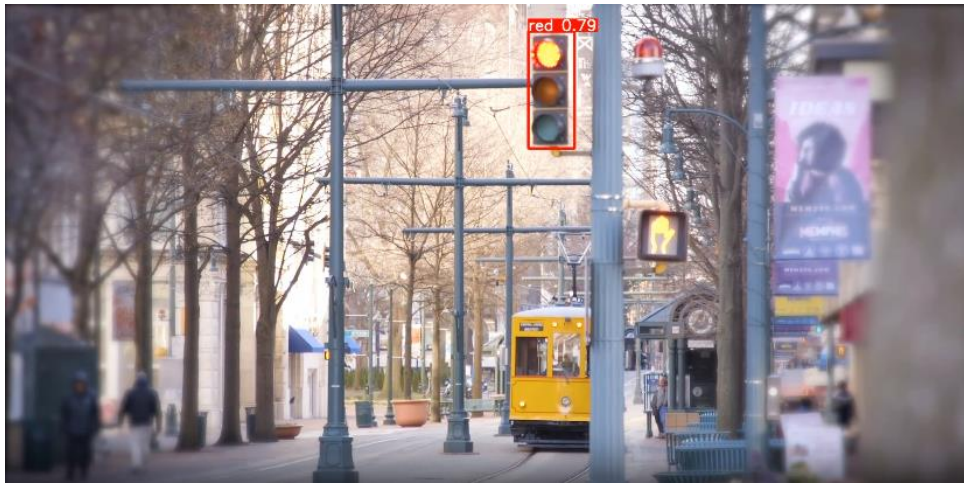


- Try to re-assemble the pieces to complete the puzzle
- Segment the various pieces from the provided images
- Use the full puzzle image as reference to get the pieces' positions
- Combine the pieces to re-create the puzzle

Already discussed in LAB3!

Project 2:
Road Signs and Traffic Lights Detection





- Detect road signs and traffic lights
- For each sign also classify it into the proper category (e.g., speed limit, stop, give way, etc..)
- Detect if the traffic light is green, yellow or red
- For speed limits read the maximum speed value

DATA FOR PROJECT2







A zip file has been provided on elearning with a few sample images (divided into an "*easy*" and an "*hard*" folder)

These two datasets can be used for further evaluation or for training ML-based approaches

- (*traffic lights*) https://zenodo.org/records/12706046 (download the two archives *dataset_train_rgb.zip* and *dataset_test_rgb.zip*)
- (*road signs*) https://www.cvl.isy.liu.se/research/datasets/traffic-signs-dataset/ (download the *Set 1*)

Algorithm to be Developed

1. Load some of the provided images
   - There are 2 sets: an "*easy*" one where the detection is relatively simple and an "*hard*" one with more challenging images
   - Links to large datasets are also provided, use them in case you decide to train ML algorithms or need large amount of data to tune your approach
2. Find the road signs and the traffic lights. You can use any object detection strategy, e.g.:
   - Deep Learning based object detectors like Yolo, R-CNN, SSD, Retina-net, etc..
   - Traditional ML approaches like Bag-of-Words or Viola and Jones
   - Also "deterministic" strategies e.g., Hough Transform for specific shapes can be used
3. For each traffic light analyze it to get if it is green, yellow or red (e.g. you can analyze the color in the different up-mid-low regions)
4. For each speed limit sign read the maximum allowed speed value (you can find the central region and use some OCR approach)

Extra Ideas (optional)

1. Try different schemes for detecting or classifying road signs and traffic light, e.g., Hough Transform, template matching
2. Generalize to different types and colors of the signs or to different types of non-standard traffic lights
3. Compare classification and OCR for reading speed limits
4. Recognize more types of road signs
5. Add provisions for night/dark environments
6. Test other images
7. Any additional idea is welcome !

Differently from HW1, this final project lab has a mark,
a few of these or any other «extra» idea could improve your mark